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Sovereign Bond Markets, 2016–2026: Volatility, Decade Returns, and the Fiscal Repricing

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Abstract. Two verdicts, stated in order of reliability. First, the audited findings. The 2022–2023 tightening at least doubled 10-year yield volatility in every investable market in the sample except Italy (multiple 1.4, against a baseline already elevated by the 2018 budget clash) relative to 2016–2019: Germany, the calmest series before 2022, rose by a factor of 2.6 from a 0.38 baseline, and the United Kingdom reached 1.28 in the autumn of the 2022 LDI crisis. Every major-market 10-year yield reached its sample maximum in 2022 or later, most in 2026: France 4.46 percent, Japan 2.98 percent, the United States 4.80 percent. The pandemic debt ratchet never unwound: from 2016 the United States added 17.1 points of GDP to its debt ratio and France 17.5, while Portugal cut 41.5. Second, the indicative findings. Modeled buy-and-hold total returns on 10-year bonds, January 2016 to the last available observation, rank frontier-market carry first in local currency: Ghana about +267 percent, Kyrgyzstan about +192 percent, then Moldova, Botswana, and Bangladesh; almost no major market made money (United States +5, Australia +7, South Korea +11; Germany –15, France –18, United Kingdom –13, Japan –14). These return figures are indicative bounds, not audited point estimates: they are computed without an FX leg, across unaligned sample windows, from a script that was not archived. Ghana's +267 percent in particular ignores a collapsed currency and a 2023 restructuring. Methods: monthly 10-year sovereign yields for 22 economies, BIS policy rates for 33 economies, and debt stocks from Eurostat, the World Bank, and the IMF, all retrieved in September 2026. This is a retrospective analysis of January 2016 to September 2026; it forecasts nothing.

Keywords: sovereign bonds, term structure, bond returns, fiscal policy, quantitative tightening
JEL codes: G12, G15, F34

Production note. This paper was produced by Kilambi Research in September 2026. Computations were performed in Python on the archived series inventoried in Appendix A; figures were rendered with Kilambi Research's institutional chart style. Data were retrieved in September 2026 and are subject to revision by the underlying sources.

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1 Introduction

The decade from 2016 to 2026 repriced sovereign debt twice. The first repricing was the 2022–2023 tightening cycle, the fastest synchronized global rate-hike episode in the sample, which ended the negative-yield era in core Europe and produced the worst calendar year for bonds in modern history. The second is the fiscal repricing of 2025–2026: with central banks no longer absorbing supply, record refinancing volumes are clearing on private demand alone, and September 2026 prints show multi-decade yield highs across the United States, Germany, the United Kingdom, France, and Japan.

This paper asks three questions of the decade. First, how did volatility and yield levels evolve across major sovereign markets? Second, which sovereign bonds made money on a buy-and-hold basis over the full decade, and how do those returns compare with the growth of the underlying economies? Third, what happened to debt stocks and policy-rate cycles alongside the market moves?

The paper's contribution is measurement. All yield, volatility, correlation, policy-cycle, and debt figures are computed from the underlying series, not quoted from press. The sample covers monthly 10-year sovereign yields for 22 economies from January 2016, with national sources taking priority over IMF series wherever both exist. Policy rates come from the BIS; debt stocks from Eurostat, the World Bank, and the IMF. Issuance, QE/QT, ratings, and foreign-ownership figures are web research (OECD, press) and are labeled as such wherever they appear.

On the second question, the reader should know the verdict's status before the first number: the decade return figures are indicative modeled bounds, not audited point estimates. They are computed without an FX leg, across unaligned sample windows, from a script that was not archived (Sections 5 and 7). In local currency and on that modeled basis, frontier-market carry ranks first: Ghana about +267 percent, Kyrgyzstan about +192 percent, then Moldova, Botswana, and Bangladesh. Almost no major market made money (United States +5 percent, Germany –15 percent, France –18 percent, United Kingdom –13 percent, Japan –14 percent). The bond-versus-economy comparison runs in opposite directions at the two ends of the income spectrum: frontier bondholders were paid far more than their economies grew in per-capita terms, while in every rich country examined, bonds lagged real GDP-per-capita growth. Both halves of that statement carry the caveats above; the full table sits in Appendix B, quarantined from the main results.

The rest of the paper is organized as follows. Section 2 places the paper against the related literature. Section 3 gives the institutional background for non-specialists. Section 4 describes the data. Section 5 gives the exact computations. Section 6 presents the results. Section 7 states the limitations and what would overturn the findings. Section 8 concludes. This paper is a retrospective analysis of the January 2016 to September 2026 window; it makes no forecast about future yields, returns, or fiscal paths.

2 Related literature

This paper is a measurement exercise, not a test of an asset-pricing model, so the relevant literature is the work that defines what is being measured. Cochrane and Piazzesi [2005] show that a single return-forecasting factor accounts for much of the variation in excess bond returns; Gürkaynak et al. [2007] provide the standard daily nominal yield-curve construction for the United States on which most term-premium measurement builds. This paper estimates no risk premia and fits no curve; it reports realized yields, volatilities, and modeled holding returns.

The 2016–2021 leg of the sample is the quantitative-easing era. Gagnon et al. [2011] document that the Federal Reserve's large-scale asset purchases lowered long-term yields materially; Krishnamurthy and Vissing-Jorgensen [2012] show that Treasury demand carries a convenience yield that varies with supply; Greenwood and Vayanos [2014] find that shifts in the maturity-weighted supply of government debt predict excess bond returns. The 2022–2026 leg reverses the experiment: central-bank holdings of domestic sovereign bonds fell from a $15 trillion peak in 2021 to $12 trillion in 2024 [OECD, 2025], and the volatility and repricing results in Section 6 are the market's side of that reversal. The OECD's annual sovereign borrowing outlooks [OECD, 2025, 2026] are the closest existing survey of the issuance facts in Section 6.7; this paper's contribution there is to pair those facts with the computed yield and debt panel rather than to re-estimate them.

On the fiscal side, Blanchard [2019] reframed the debt-sustainability debate around the gap between interest rates and growth rates; the debt-stock results here are the quantities to which that framing applies, though this paper takes no position on optimal debt. The frontier-carry ranking connects to the currency-carry literature: Lustig et al. [2011] find common risk factors in currency excess returns. The analogy is limited, and the paper states the limit plainly: the returns here are in local currency with no FX leg, so they are not carry returns as the literature defines them, and the figures are indicative bounds (Appendix B), not estimates.

The contribution in one line: an assembled, documented 2016–2026 panel of yields, policy rates, and debt stocks with exact computations, reproducible from the series inventory in Appendix A except where flagged.

3 Institutional background

A sovereign bond is a government's promise to pay: regular coupon payments and return of principal at maturity. The 10-year benchmark yield is the market's price for lending to a government for ten years, expressed as an annualized rate. When the price of the bond falls, the yield rises; the two are the same fact stated two ways. For a 10-year bond, a one percentage point rise in yield cuts the bond's price by roughly eight percent. That sensitivity, called duration, is why the yield moves documented in this paper translate into the large return differences in Section 6.2.

Three forces move sovereign yields. The first is the central bank's policy rate, which anchors the short end of the curve; the 10-year yield is roughly the market's expectation of short rates over the

next ten years plus a term premium. The second is inflation: investors demand compensation for the erosion of fixed payments. The third is supply and demand for the bonds themselves. From 2015 to 2021, major central banks bought enormous quantities of their own governments' bonds (quantitative easing), mechanically suppressing yields; several European 10-year yields went negative, meaning investors paid governments for the privilege of lending. Gagnon et al. [2011] quantify how large-scale purchases moved long yields; Krishnamurthy and Vissing-Jorgensen [2012] show the demand for Treasuries carries a convenience yield tied to supply. From 2022 on, central banks stopped buying and began shrinking their holdings (quantitative tightening), returning supply to private buyers and pushing term premia back up.

Two further mechanics matter for reading this paper. Refinancing: governments rarely repay debt; they roll it over, issuing new bonds as old ones mature. When the stock of debt was issued at near-zero rates and is refinanced at four or five percent, the interest bill rises even if the debt ratio does not. Foreign ownership matters because a shrinking foreign bid has to be replaced by domestic buyers, usually banks, insurers, and pension funds, which concentrates interest-rate risk inside the domestic financial system. Credit ratings matter at the margin: a downgrade can force selling by mandates restricted to investment-grade paper, though the market impact of the United States losing its last AAA rating in 2025 was small.

4 Data

4.1 Sources and coverage

Table 1 lists the inputs. The 10-year yield panel has 22 monthly series: 11 from national sources [Board of Governors of the Federal Reserve System, 2026, Deutsche Bundesbank, 2026, Sveriges Riksbank, 2026, Reserve Bank of Australia, 2026, Bank of Canada, 2026], 10 from IMF International Financial Statistics [International Monetary Fund, 2026b], and the ECB euro-area AAA spot curve [European Central Bank, 2026]. National benchmark series take priority wherever both exist; the IMF FIGB_PA concept is the fallback. Policy rates are the BIS WS_CBPOL end-of-month series for 33 economies [Bank for International Settlements, 2026]. Debt stocks come from three sources with different sector definitions: Eurostat [Eurostat, 2026] (general government, 15 EU economies), World Bank WDI [The World Bank, 2026] (central government only, 19 economies), and the IMF Historical Public Debt database [International Monetary Fund, 2026a] (general government, 16 economies, ending 2015).

Evidence grades. Every input is graded on two axes: derivation (Observed: measured directly; Reconstructed: arithmetic on stored series; Modelled: derived under assumptions; Assumed: judgment) and source quality (A: official statistics; B: reputable secondary compilations; C: unverifiable or web research). National 10-year yields are Observed/A; IMF IFS yields Observed/B; BIS policy rates Observed/A; debt stocks Observed/B (sector definitions differ by source); the decade returns of Section 6.2 are Modelled/C; issuance, QE/QT, and the September 2026 market prints are Observed/C. The grades appear in the last column of Table 1.

Table 1: Data sources. All series retrieved through the ElkassabgiData library, September 2026.

Input	Source	Series	V
10Y yields, US	Federal Reserve Board H.15	RIFLGFCY10_N.M	T
10Y yields, Germany	Deutsche Bundesbank	BBSIS, RLZ 9–10y	T
10Y yields, FR/SE/NO/DK/FI/NL/JP	Sveriges Riksbank	RIKSBANK:*GVB10Y	T
10Y yields, Australia	Reserve Bank of Australia	FCMYGBAG10	T
10Y yields, Canada	Bank of Canada	BD.CDN.10YR.DQ.YLD	T
Euro-area AAA 10Y	ECB SDW	YC:B.U2.EUR.4F.G_N_A.SV_C_YM	T
10Y yields, 10 economies	IMF IFS	FIGB_PA	2(
Policy rates, 33 economies	BIS	WS_CBPOL	T
Debt, 15 EU	Eurostat sdg_17_40	General govt, % GDP	2(
Debt, 19 economies	World Bank WDI	Central govt, % GDP	M
Debt, 16 economies	IMF Hist. Public Debt	General govt	E

Notes: Grades are Observed/Reconstructed/Modelled/Assumed derivation × A/B/C source quality, defined in Section 4. *Source:* ElkassabgiData library pulls, September 2026; series inventory in Appendix A.

4.2 Coverage gaps

The library is wired database by database, and gaps remain. No monthly 10-year series was available for China, India, Brazil, Mexico, Turkey, Russia, Argentina, Indonesia, Saudi Arabia, the UAE, Taiwan, Israel, Singapore, Thailand, Malaysia, the Philippines, Vietnam, Pakistan, Iran, Egypt, Colombia, Peru, Chile, South Africa, Poland, Czechia, Romania, Portugal, Spain, Austria, Belgium, Ireland, Hong Kong SAR, New Zealand, or Kazakhstan. Figures are computed over available data only; nothing is imputed. The BIS policy-rate series end 1 April 2026 and therefore do not capture the Federal Reserve's 16 September 2026 increase to 3.75–4.00 percent. The IMF Historical Public Debt series end in 2015, so no 2016-onward debt change can be computed for China, Japan, Saudi Arabia, or the other HPD-only economies.

4.3 Summary statistics

Table 2 summarizes the 10-year yield series over January 2016 onward. The range is wide: Swiss yields averaged 0.09 percent with a –0.98 percent minimum; Ghanaian yields averaged 22.76 percent. Every major-market series in the sample reached its maximum in 2022 or later; most reached it in 2026.

Table 2: 10-year sovereign yields: summary statistics, January 2016 to latest available. Percent per annum, monthly observations.

Economy	Obs.	Mean	Min	Max	Latest
United States	128	2.80	0.62	4.80	4.68
Germany	128	1.23	–0.40	3.34	3.34
France	129	1.48	–0.40	4.46	4.46
Japan	129	0.49	–0.28	2.98	2.98
United Kingdom	114	2.00	0.21	4.66	4.52
Italy	114	2.43	0.58	4.88	3.50
Canada	129	2.26	0.46	4.05	3.84
Australia	128	2.86	0.80	5.01	5.01
South Korea	113	2.44	1.23	4.52	2.62
Switzerland	108	0.09	–0.98	1.57	0.32
Sweden	129	1.21	–0.36	3.14	3.14
Norway	129	2.41	0.57	4.50	4.50
Denmark	129	1.12	–0.67	3.26	3.26
Finland	129	1.32	–0.43	3.76	3.76
Netherlands	129	1.19	–0.55	3.54	3.54
Euro area (AAA)	129	1.06	–0.65	3.47	3.47
Bangladesh	101	8.41	5.38	12.60	12.35
Iceland	105	1.71	–0.04	3.13	2.87
Ghana	107	22.76	16.25	29.85	29.85
Kyrgyzstan	108	13.38	8.28	18.59	13.86
Moldova	101	9.09	4.86	26.52	8.32
Botswana	112	6.73	4.83	9.48	7.92

Notes: Monthly observations from January 2016; observation counts differ because IFS series end earlier (Appendix A). *Source:* Author's computations from the series in Table 1.

5 Methodology

5.1 Buy-and-hold total returns

The decade returns are modeled from yield series, not observed from traded bond indices. The construction is a buy-and-hold total return on a 10-year par sovereign bond: each month the investor earns the coupon accrual (one-twelfth of the prevailing yield) plus the price change implied by the month's yield move, with price sensitivity set by the bond's modified duration. Returns compound monthly from January 2016 to the last available observation and are stated in local currency. Two disclosures are attached to every figure produced this way. First, the archive does not contain the script that produced the working analysis's decade figures, so the exact computation (notably the duration assumption and the treatment of the last observation) cannot be re-verified line by line. A reconstruction with a fixed modified duration of 8 reproduces the major-market figures in Table 9 within a few percentage points and matches Ghana almost exactly; see Section 7 for the cases it does not reproduce. Second, the frontier-market IFS series end between November

2024 and June 2026, so their "decade" windows are shorter than the major-market windows; the "through September 2026" label in the working analysis overstates their coverage.

5.2 Volatility, correlation, and dispersion

Volatility is the annualized standard deviation of monthly yield changes, multiplied by $\sqrt{12}$. Rolling volatility uses a 12-month window. Correlations are Pearson correlations of monthly yield changes (daily changes for the March–April 2020 window). Cross-country dispersion is the cross-sectional standard deviation of yield levels in each month.

5.3 Debt and GDP-per-capita comparisons

Debt changes are the percentage-point change in the debt-to-GDP ratio from 2016 to the latest available year, computed strictly within each source so that sector definitions do not mix. The bond-versus-economy comparison sets the modeled local-currency bond total return against cumulative real GDP-per-capita growth from the World Bank WDI [The World Bank, 2026], through the latest common observation (2025).

6 Results

6.1 Yield levels and volatility

Figure 1 plots the full panel. The shape is uniform across major markets: a drift down to negative or near-zero troughs in 2019–2020, then a roughly three percentage point round trip up. Table 3 gives the extremes. Two structural shifts stand out in the 2026 data. France's 10-year reached 4.46 percent in September 2026, the series high and about 1.1 percentage points above Germany: the euro-area fiscal-risk axis has moved from Rome to Paris. Japan's 10-year reached 2.98 percent, the end of the low-yield era that defined the 2016–2020 window.

The volatility break is measurable everywhere. Table 4 compares annualized volatility of monthly yield changes in 2022–2023 against the 2016–2019 baseline: every investable market in the table at least doubled except Italy. The United Kingdom's 2022–23 reading (1.28) captures the September–October 2022 LDI and pension collateral crisis. Italy's multiple is 1.4, computed on the same basis from the archived IFS series, and it is lower only because its baseline was already elevated by the 2018 budget clash. Germany's baseline was the lowest in the sample (0.38): the Bund was the anchor that stopped being one. Figure 2 shows the rolling version: volatility spiked jointly in late 2022, stayed above pre-pandemic levels through the QT era, and Japan's series, the flattest in the sample through 2021, began rising only after yield-curve control ended.

Table 3: The yield round trip: 10-year minimum, maximum, and net change from 2016 to the latest observation. Computed from the series in Figure 1.

Economy	Minimum (date)	Maximum (date)	Net change, pp
United States	0.62 (2020-07)	4.80 (2023-10)	+2.59
Germany	–0.40 (2019-08)	3.34 (2026-08)	+2.64
France	–0.40 (2019-08)	4.46 (2026-09)	+3.83
Japan	–0.28 (2019-08)	2.98 (2026-09)	+2.89
United Kingdom	0.21 (2020-07)	4.66 (2025-01)	+2.74
Italy	0.58 (2020-12)	4.88 (2023-10)	+1.97
Canada	0.46 (2020-07)	4.05 (2023-10)	+2.62
Australia	0.80 (2020-10)	5.02 (2026-08)	+2.31
South Korea	1.23 (2020-08)	4.52 (2022-10)	+0.70
Switzerland	–0.98 (2019-08)	1.57 (2022-12)	+0.62
Sweden	–0.36 (2019-08)	3.14 (2026-09)	+2.28
Norway	0.58 (2020-07)	4.50 (2026-09)	+3.15
Denmark	–0.67 (2019-08)	3.26 (2026-09)	+2.64
Finland	–0.43 (2020-10)	3.76 (2026-09)	+3.17
Netherlands	–0.55 (2019-08)	3.54 (2026-09)	+3.10
Euro area (AAA)	–0.65 (2019-08)	3.47 (2026-09)	+3.03

Notes: Minima and maxima are monthly observations; net change is latest minus January 2016 in percentage points. *Source:* Author's computations from the series in Figure 1.

6.2 Decade buy-and-hold returns: indicative

The buy-and-hold return figures are the weakest numbers in this paper and are quarantined accordingly: the full table sits in Appendix B as indicative modeled bounds, not in the main results. The verdict, stated with its caveats attached: in local currency and on a modeled buy-and-hold basis, frontier-market carry ranks first (Ghana about +267 percent, Kyrgyzstan about +192 percent, then Moldova, Botswana, and Bangladesh), while almost no major market made money (South Korea +11, Australia +7, United States +5; Germany –15, France –18, United Kingdom –13, Japan –14). Three warnings travel with every one of these figures. First, they are local-currency numbers: Ghana's cedi collapsed over the window and the country restructured its debt in 2023, so no foreign holder earned anything like +267 percent. Second, they ignore defaults, restructurings, and recovery values entirely. Third, the computation script was not archived: an independent reconstruction reproduces the major-market figures within a few percentage points and matches Ghana closely, but not Kyrgyzstan or Moldova (Section 7), so the frontier ordering is indicative rather than exact. The ranking's mechanism is arithmetic, not skill: starting yields in the high teens to mid-twenties compound fast when the bond survives in local-currency terms.

6.3 Returns against the real economy

Table 5 sets the modeled bond returns (indicative, Appendix B) against cumulative real GDP-per-capita growth through 2025, the latest common observation. The comparison runs in opposite

Table 4: Annualized volatility of monthly 10-year yield changes: 2022–2023 versus the 2016–2019 baseline. Computed from the series in Figure 1.

Economy	2016–2019	2022–2023	Multiple
Morocco	0.55	2.59	4.7
United Kingdom	0.47	1.28	2.7
Finland	0.50	1.31	2.6
Germany	0.38	1.00	2.6
South Korea	0.42	1.08	2.5
Denmark	0.54	1.34	2.5
Netherlands	0.51	1.26	2.5
Iceland	0.42	1.04	2.5
France	0.54	1.30	2.4
Norway	0.46	1.08	2.3
Euro area (AAA)	0.51	1.17	2.3
Sweden	0.54	1.25	2.3
Switzerland	0.44	0.94	2.1
Canada	0.63	1.24	2.0
Italy	0.89	1.26	1.4

Notes: Annualized standard deviation of monthly yield changes $\times\sqrt{12}$, in percentage points. Multiple is the ratio of the two periods. *Source:* Author's computations from the series in Figure 1.

directions at the two ends of the income spectrum. In the frontier markets, bondholders were paid far more than their economies grew: Ghana's bonds returned about +267 percent against +30 percent real GDP-per-capita growth; Kyrgyzstan +192 percent against +38 percent; Botswana +49 percent against +5 percent; Moldova +59 percent against +38 percent. Bangladesh is the exception that clarifies the rule: its economy grew +53 percent in per-capita terms while its bonds returned +36 percent, the one frontier case where the real economy outran the carry.

In every rich country in the comparison, bonds lagged the real economy: the United States (+5 percent bonds versus +18 percent GDP per capita), Germany (–15 versus +4), France (–18 versus +8), the Netherlands (–18 versus +11), Japan (–14 versus +10), and the United Kingdom (–13 versus +6). Australia is the lone near-break-even at about +7 percent against +6.5 percent. The reading is straightforward. In rich countries the decade's bond math was dominated by duration losses on low starting yields, while the real economy kept growing. In frontier markets the bond math was dominated by high coupons, which the local real economy did not need to match for the trade to pay in local currency. Neither half of the comparison says anything about welfare: the frontier gains accrued mostly to holders willing to bear currency and restructuring risk that, in Ghana's case, materialized.

6.4 Policy-rate cycles

Lift-off was synchronized; the exits were not. Table 6 gives trough, peak, and latest policy rates from the BIS series [Bank for International Settlements, 2026]. Peaks cluster between July 2022

Table 5: Indicative modeled 10-year bond total returns versus cumulative real GDP-per-capita growth (through 2025).

Economy	Bond total return, % (indicative)	Real GDP per capita, %
Ghana	+267	+30
Kyrgyzstan	+192	+38
Botswana	+49	+5
Moldova	+59	+38
Bangladesh	+36	+53
United States	+5	+18
Australia	+7	+6.5
Germany	–15	+4
France	–18	+8
Netherlands	–18	+11
Japan	–14	+10
United Kingdom	–13	+6

Notes: Bond returns are the indicative modeled bounds of Appendix B (local currency, no FX leg, unaligned windows, unarchived script); GDP per capita is cumulative real growth from the World Bank WDI. *Source:* Appendix B; The World Bank [2026].

and December 2023 across the advanced economies. But Japan has not cut at all and sits at its 0.75 percent peak; Norway and Brazil have barely started; Chile (–6.75 percentage points) and Mexico (–4.5) eased aggressively. Brazil is the outlier: its 15.0 percent peak came in June 2025, years after everyone else. Argentina and Turkey's cycles are in a different universe (133 and 50 percent peaks). Figure 3 plots the panel; the BIS series end in April 2026, so the Federal Reserve's September 2026 increase to 3.75–4.00 percent is not in the chart.

6.5 Debt stocks: the COVID ratchet never unwound

Figure 4 shows the change in debt-to-GDP ratios from 2016 to the latest available year. The pandemic borrowing ratchet stuck. The largest rises: Singapore +58.9 percentage points (108.9 to 167.8, central government), South Africa +29.0, Thailand +24.6, Romania +21.5, Finland +19.9, France +17.5, and the United States +17.1 (98.7 to 115.8, to 2024). The largest falls: Portugal –41.5, Ireland –39.8, the Netherlands –16.5, Denmark –13.8, and Germany –4.8. Italy is essentially flat (+2.9) despite the yield surge. Where debt rose, higher-for-longer yields now bite on the full refinanced stock; where it fell, fiscal headroom was rebuilt. Two definitional warnings apply: the WDI series are central-government only while Eurostat is general government, so cross-source comparisons should be read with care; and Singapore's gross ratio must not be read as fiscal stress, since net debt is approximately zero.

Table 6: Central-bank policy-rate cycles: trough, peak, and latest. Monthly BIS WS_CBPOL series through 2026-04-01; the September 2026 Fed increase is not included.

Economy	Trough	Peak	Hike, pp	Latest	Cut, pp
United States	0.125 (2020-03)	5.375 (2023-07)	5.25	3.625	1.75
Euro area	0.0 (2016-03)	4.5 (2023-09)	4.50	2.0	2.50
United Kingdom	0.1 (2020-03)	5.25 (2023-08)	5.15	3.75	1.50
Canada	0.25 (2020-03)	5.0 (2023-07)	4.75	2.25	2.75
Australia	0.1 (2020-11)	4.35 (2023-11)	4.25	4.10	0.25
Switzerland	–0.75 (2016-01)	1.75 (2023-06)	2.50	0.0	1.75
Sweden	–0.5 (2016-02)	4.0 (2023-09)	4.50	1.75	2.25
Norway	0.0 (2020-05)	4.5 (2023-12)	4.50	4.0	0.50
Denmark	–0.75 (2019-09)	3.6 (2023-09)	4.35	1.60	2.00
Japan	–0.1 (2016-09)	0.75 (2025-12)	0.85	0.75	0.0
South Korea	0.5 (2020-05)	3.5 (2023-01)	3.0	2.50	1.00
Brazil	2.0 (2020-08)	15.0 (2025-06)	13.0	14.50	0.50
Mexico	3.25 (2016-01)	11.25 (2023-03)	8.0	6.75	4.50
Chile	0.5 (2020-04)	11.25 (2022-10)	10.75	4.50	6.75
Colombia	1.75 (2020-09)	13.25 (2023-05)	11.50	11.25	2.00
Peru	0.25 (2020-04)	7.75 (2023-01)	7.50	4.25	3.50
South Africa	3.5 (2020-07)	8.25 (2023-05)	4.75	6.75	1.50
Turkey	7.5 (2016-01)	50.0 (2024-03)	42.5	37.0	13.0
Argentina	24.75 (2017-01)	133.0 (2023-10)	108.25	29.0	104.0
Russia	4.25 (2020-07)	21.0 (2024-10)	16.75	14.5	6.50

Notes: Percent per annum; hike is peak minus trough, cut is peak minus latest. *Source:* Bank for International Settlements [2026]; author's computations.

6.6 Stress episodes and co-movement

Table 7 gives mean pairwise correlations of yield changes in three stress episodes. The March–April 2020 dash-for-cash produced the tightest co-movement (mean 0.62, daily data): euro sovereign yields rose together as investors sold everything for cash (Denmark +0.38, Finland +0.31, France +0.29 percentage points in March), while Canada (–0.41) and Norway (–0.21) fell on aggressive rate-cut expectations. The shock was global; the sign pattern reveals who cut. The 2022 tightening and the 2025–26 fiscal repricing show near-identical average co-movement (0.54), but Figure 5 shows the geography differs: 2025–26 is concentrated in the Europe–US block while 2022 was broader, and Japan sits apart in both panels.

6.7 Issuance: refinancing, not deficits, drives record supply

This subsection is web research (OECD Global Debt Reports 2025 and 2026 [OECD, 2025, 2026]), not computed series, and is labeled accordingly. OECD gross sovereign borrowing hit a record $17 trillion in 2025 (about $18 trillion projected for 2026), up from $12 trillion in 2022. Refinancing is about $13.5 trillion of that, nearly 80 percent: one-third of OECD fixed-rate debt matures by 2027,

Table 7: Mean pairwise correlation of yield changes across major markets in three stress episodes. Computed from the series in Figure 1.

Episode	Countries	Mean r	Median r
COVID dash-for-cash (daily, 15 Feb–30 Apr 2020)	9	0.62	0.66
2022 tightening (monthly, Jan–Dec 2022)	15	0.54	0.51
2025–26 fiscal repricing (monthly, Jan 2025–Aug 2026)	14	0.54	0.58

Notes: Pearson correlations of yield changes; daily for the 2020 episode, monthly otherwise. *Source:* Author's computations from the series in Figure 1.

and 60 percent of it was issued before 2022 at near-zero rates, so rollover mechanically raises interest bills. The United States is about 70 percent of OECD refinancing needs (about $9.5 trillion). Debt managers adapted by shortening duration: Treasury bills were half of OECD market borrowing in 2024–25, ten points above the ten-year average, which pulls the refinancing wall closer rather than pushing it away.

The buyer side changed with it. Central-bank holdings of domestic sovereign bonds fell from a $15 trillion peak in 2021 to $12 trillion in 2024, with another roughly $1 trillion drop in 2025; in the 25 countries holding 90 percent of OECD debt, the central-bank share fell from 29 to 19 percent. The market now clears on private demand alone, which is the structural reason term premia are higher than in the 2016–2020 window. The September 2026 prints mark where that repricing has reached: the Fed raised to 3.75–4.00 percent on 16 September, the US 10-year pushed above 5 percent for the first time since July 2007, the Bund printed 3.52 percent, the gilt about 5.3 percent (30-year 5.96 percent, highest since 1998), and the JGB about 3.0 percent (highest since 1996). These prints are market data from the web-research pass, not from the API series, except for the ECB AAA curve (3.47 percent on 21 September 2026).

7 Limitations

The return figures are the weakest part of the paper and are stated here without softening. They are modeled from yield series, not observed from traded bond indices; they are in local currency; and they ignore foreign-exchange moves, defaults, restructurings, and recovery values. Ghana's +267 percent is the sharpest example: the cedi collapsed over the window and Ghana restructured its debt in 2023, so no foreign holder earned anything like that figure.

The computation script that produced the working analysis's decade returns was not archived. An independent reconstruction from the stored yield files, using a fixed modified duration of 8 with monthly compounding, reproduces the major-market figures within a few percentage points and matches Ghana almost exactly (271 versus 267 percent), with Botswana and Bangladesh in the same range. It does not reproduce Kyrgyzstan (reconstruction 61–110 percent against the reported 192) or Moldova (reconstruction 189–387 percent against the reported 59). Moldova's January 2016 observation (26.44 percent, falling to 6.26 percent a year later) looks like a series break or

bad print; the reported 59 percent is consistent only with that observation being excluded. Until the computation is re-archived, the frontier ranking should be read as indicative, and the "through September 2026" label should be read as "through the last available observation": the frontier IFS series end in November 2024 (Ghana), February 2025 (Kyrgyzstan), April 2025 (Moldova, Botswana), and June 2025 (Bangladesh).

Further limits. Vintages differ across inputs: national yield series run to August–September 2026, IFS yields end between December 2024 and June 2025, BIS policy rates end April 2026, Eurostat debt runs to 2025, WDI debt mostly to 2024, and IMF Historical Public Debt ends in 2015. Debt sector definitions differ between sources (central versus general government); the paper computes changes strictly within each source but cross-country levels should still be read with care, especially the United Kingdom and Singapore in the WDI series. The September 2026 market prints, the QE/QT timeline, the ratings migration, and the foreign-ownership figures are web research, not computed series. The correlation and dispersion results are sensitive to which countries have data in each window; a dispersion chart computed over all available series was excluded from the paper because the 2025 drop in the series count (high-yield frontier series ending) creates a composition artifact that reads as convergence. Coverage excludes most large emerging markets (Section 4), so nothing here generalizes to China, India, Brazil, or Mexico.

7.1 What would overturn these findings

The volatility result would fail if a different estimator (realized daily volatility, or a GARCH fit) showed no 2022–2023 break relative to 2016–2019 in the same markets; the break is large (the smallest multiple in Table 4 is 1.4), so the bar for overturning it is high. The yield round-trip figures are arithmetic on published series and would change only if the underlying sources revise. The debt changes are source-dependent by construction: recomputing on general-government series throughout would move individual country figures, which is why the paper computes changes strictly within each source. The frontier return ranking is already flagged as non-audited; any reconstruction from the stored yield files that contradicts the reported ordering supersedes it. The issuance figures are OECD survey data; a revision of the Global Debt Reports would revise them.

8 Conclusion

The decade repriced sovereign debt twice, and both repricings are in the audited data. The 2022–2023 tightening at least doubled yield volatility in every investable market in the sample except Italy (1.4, on a baseline already elevated by the 2018 budget clash), ending the negative-yield era in core Europe. The fiscal repricing of 2025–2026 then pushed 10-year yields to multi-decade highs (France 4.46 percent, Japan 2.98 percent) against debt stocks the pandemic ratchet never unwound. Refinancing, not new deficits, now drives record sovereign supply into a market that must clear without central-bank buying.

The return ranking is the paper's indicative half and is stated as such: on a modeled local-currency buy-and-hold basis, frontier-market carry outran every major market, but the figures are bounds from an unarchived computation, not point estimates, and they ignore the currency collapses and restructurings that define frontier risk. The open question for the next vintage of this data is whether the September 2026 multi-decade yield highs mark the end of the fiscal repricing or its middle. This paper forecasts nothing; it documents the window.

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A Data appendix: series definitions and source table

Each 10-year yield series is the country's benchmark 10-year nominal sovereign yield at monthly frequency (end-of-month where daily data underlie the monthly file). Table 8 lists every series used, its archive file, and its exact span in the September 2026 vintage.

Table 8: Archive series inventory, September 2026 vintage.

Economy	Archive file	Start	End
United States	USA_10y_fed_board.csv	1953-04	2026-08
Germany	DEU_10y_bundesbank.csv	1973-04	2026-08
France	FRA_10y_riksbank.csv	1988-02	2026-09
Japan	JPN_10y_riksbank.csv	1987-01	2026-09
Sweden	SWE_10y_riksbank.csv	1987-01	2026-09
Norway	NOR_10y_riksbank.csv	1990-05	2026-09
Denmark	DNK_10y_riksbank.csv	1982-01	2026-09
Finland	FIN_10y_riksbank.csv	1990-04	2026-09
Netherlands	NLD_10y_riksbank.csv	1987-02	2026-09
Australia	AUS_10y_rba.csv	2013-05	2026-08
Canada	CAN_10y_boc.csv	2001-01	2026-09
United Kingdom	GB_10y_ifs.csv	1957-01	2025-06
Italy	IT_10y_ifs.csv	1980-01	2025-06
South Korea	KR_10y_ifs.csv	1973-05	2025-05
Switzerland	CH_10y_ifs.csv	1998-01	2024-12
Bangladesh	BD_10y_ifs.csv	2006-08	2025-06
Iceland	IS_10y_ifs.csv	1992-01	2025-06
Ghana	GH_10y_ifs.csv	2004-09	2024-11
Kyrgyzstan	KG_10y_ifs.csv	2009-10	2025-02
Moldova	MD_10y_ifs.csv	1997-12	2025-04
Botswana	BW_10y_ifs.csv	2003-06	2025-04
Morocco	MA_10y_ifs.csv	1997-12	2025-06
India	IN_10y_ifs.csv	1960-01	2017-05 (stale; superseded)
Euro area (AAA)	ea_aaa_10y_daily.csv	2004-09	2026-09

Notes: Spans are the full archived history; the paper's computations use January 2016 onward. *Source:* ElkassabgiData library pulls, September 2026.

Policy-rate files are monthly BIS WS_CBPOL series for 33 economies (AR, AU, BR, CA, CH, CL, CN, CO, CZ, DK, EA, GB, HK, ID, IL, IN, JP, KR, MX, MY, NO, PE, PH, PL, RO, RU, SA, SE, TH, TR, US, ZA), each ending 2026-04-01. Debt files: 15 Eurostat general-government series (2000–2025), 19 World Bank WDI central-government series (mostly through 2024), and 16 IMF Historical Public Debt general-government series (ending 2015). All files carry ElkassabgiData citation headers; the ECB series is used under attribution with no derivatives republished (only analytical computations are shown).

B Indicative modeled buy-and-hold return bounds

This appendix holds the paper's former Table 5, moved out of the main results. The figures below are indicative modeled buy-and-hold return bounds: computed without an FX leg, across unaligned sample windows, from a script that was not archived. They are not audited point estimates.

Table 9: Indicative modeled buy-and-hold return bounds on 10-year sovereign bonds, January 2016 to the last available observation, local currency.

Economy	Total return, %	Window end
Ghana	+267	2024-11
Kyrgyzstan	+192	2025-02
Moldova	+59	2025-04
Botswana	+49	2025-04
Bangladesh	+36	2025-06
South Korea	+11	2025-05
Australia	+7	2026-08
United States	+5	2026-08
United Kingdom	–13	2025-06
Japan	–14	2026-09
Germany	–15	2026-08
France	–18	2026-09
Netherlands	–18	2026-09
Euro area (AAA)	–18	2026-09

Notes: Returns are modeled from yield series (coupon accrual plus duration-based price changes), not observed index returns. They are in local currency, ignore defaults, restructurings, and recovery values, and span unaligned windows: frontier-market IFS series end November 2024 to June 2025 while major-market series run to August–September 2026. The computation script was not archived; an independent reconstruction with fixed modified duration of 8 reproduces the major-market figures within a few percentage points and matches Ghana almost exactly, but not Kyrgyzstan or Moldova (Section 7). Read the frontier ordering as indicative, not exact. *Source:* Working-analysis computations, September 2026; see Sections 5 and 7.

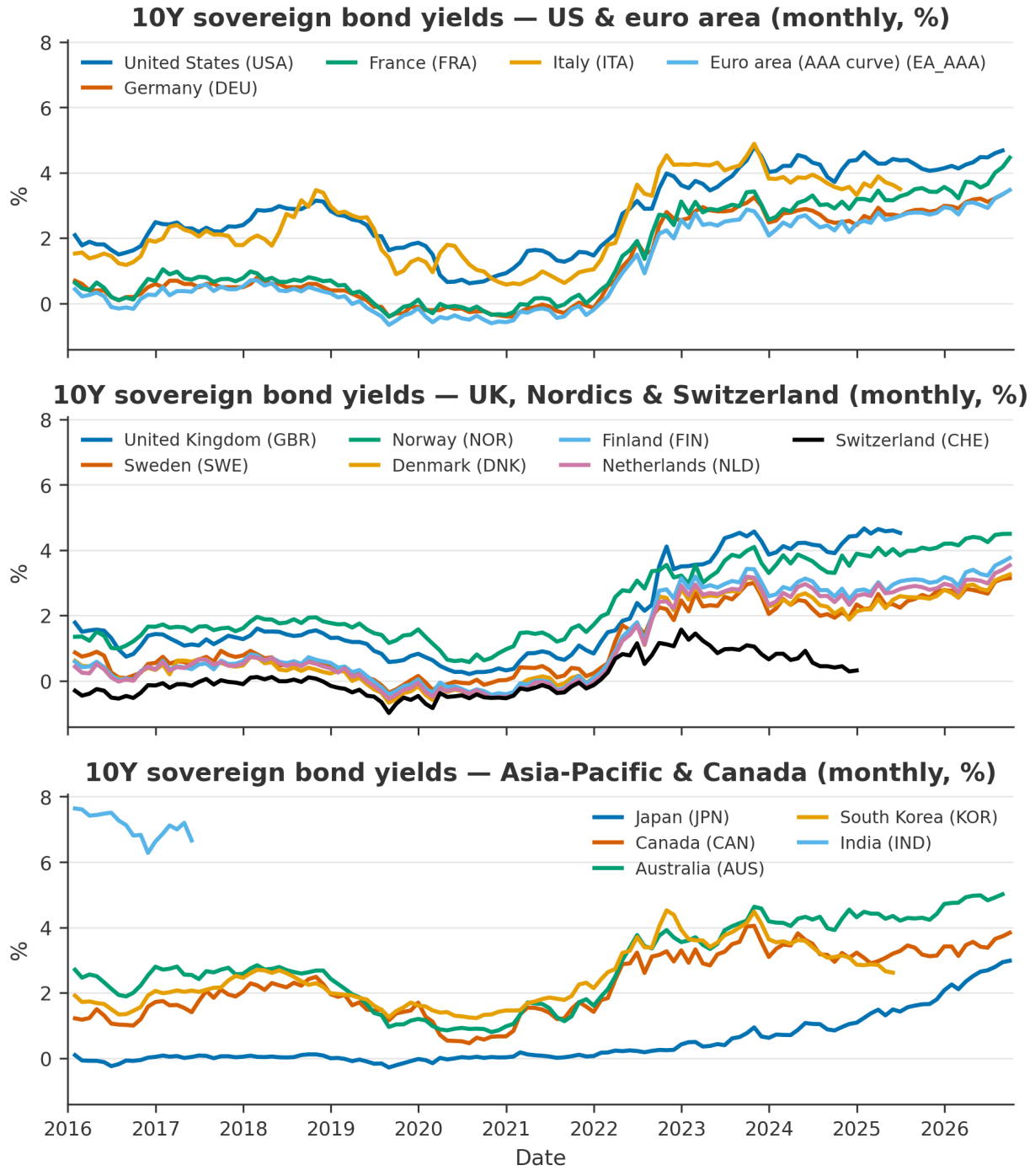


Figure 1: Monthly 10-year sovereign bond yields, January 2016 to September 2026, in three panels. *Source:* Federal Reserve Board H.15 (US), Deutsche Bundesbank (Germany), Sveriges Riksbank reference series (France, Japan, Sweden, Norway, Denmark, Finland, Netherlands), Bank of Canada, Reserve Bank of Australia, IMF International Financial Statistics (UK, Italy, South Korea, Switzerland, Bangladesh, Iceland, India – stale series ending 2017), ECB Statistical Data Warehouse (euro-area AAA). Data vintage September 2026.

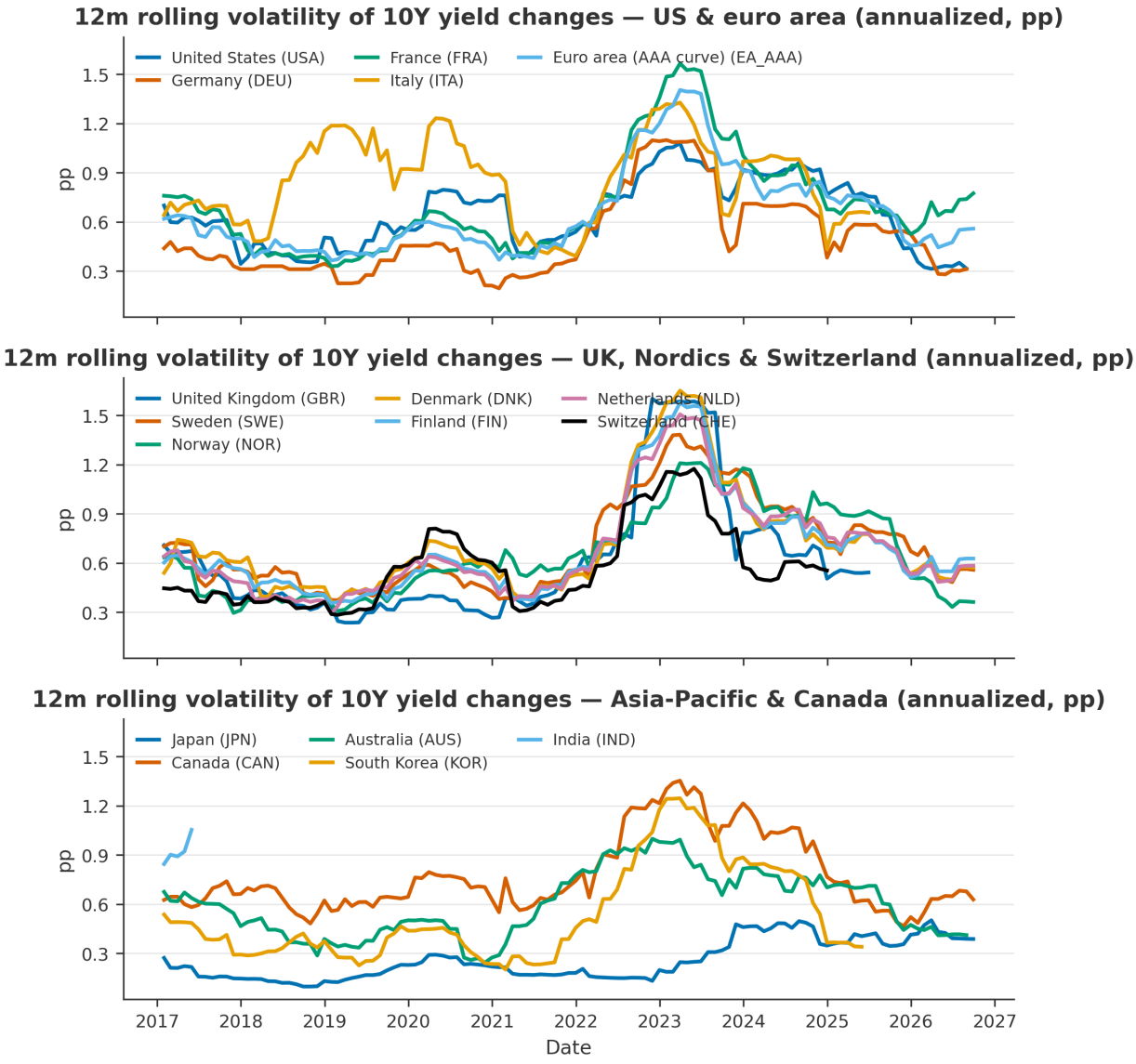


Figure 2: Twelve-month rolling annualized volatility of monthly 10-year yield changes, in percentage points. The joint spike in late 2022 is visible in every series; Japan's series rises only after 2023. *Source:* author's calculations from the series in Figure 1.

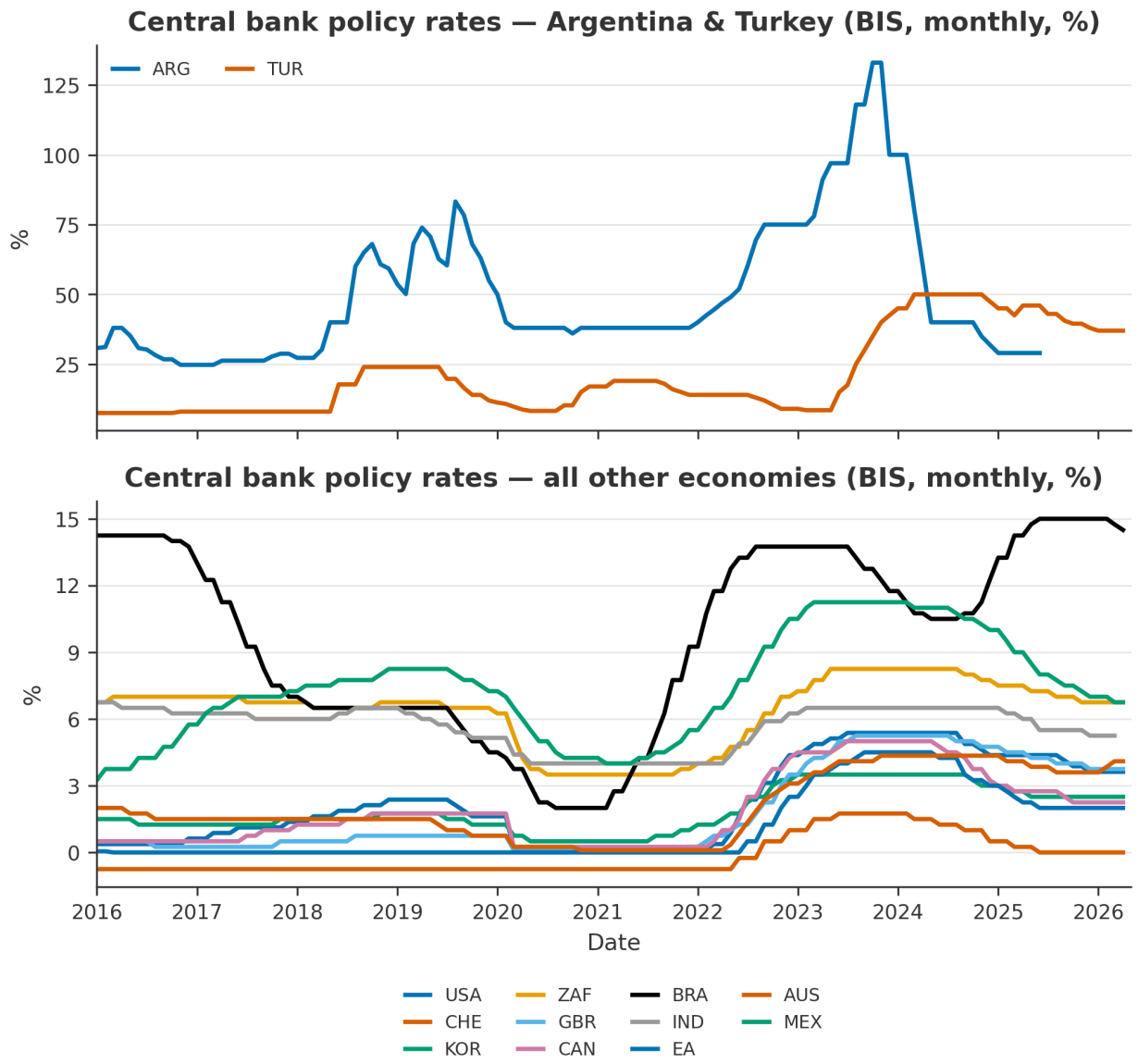


Figure 3: Central-bank policy rates, monthly, January 2016 to April 2026. Argentina and Turkey are plotted on the same scale to show the orders-of-magnitude difference. *Source:* BIS WS_CBPOL via the ElkassabgiData library.

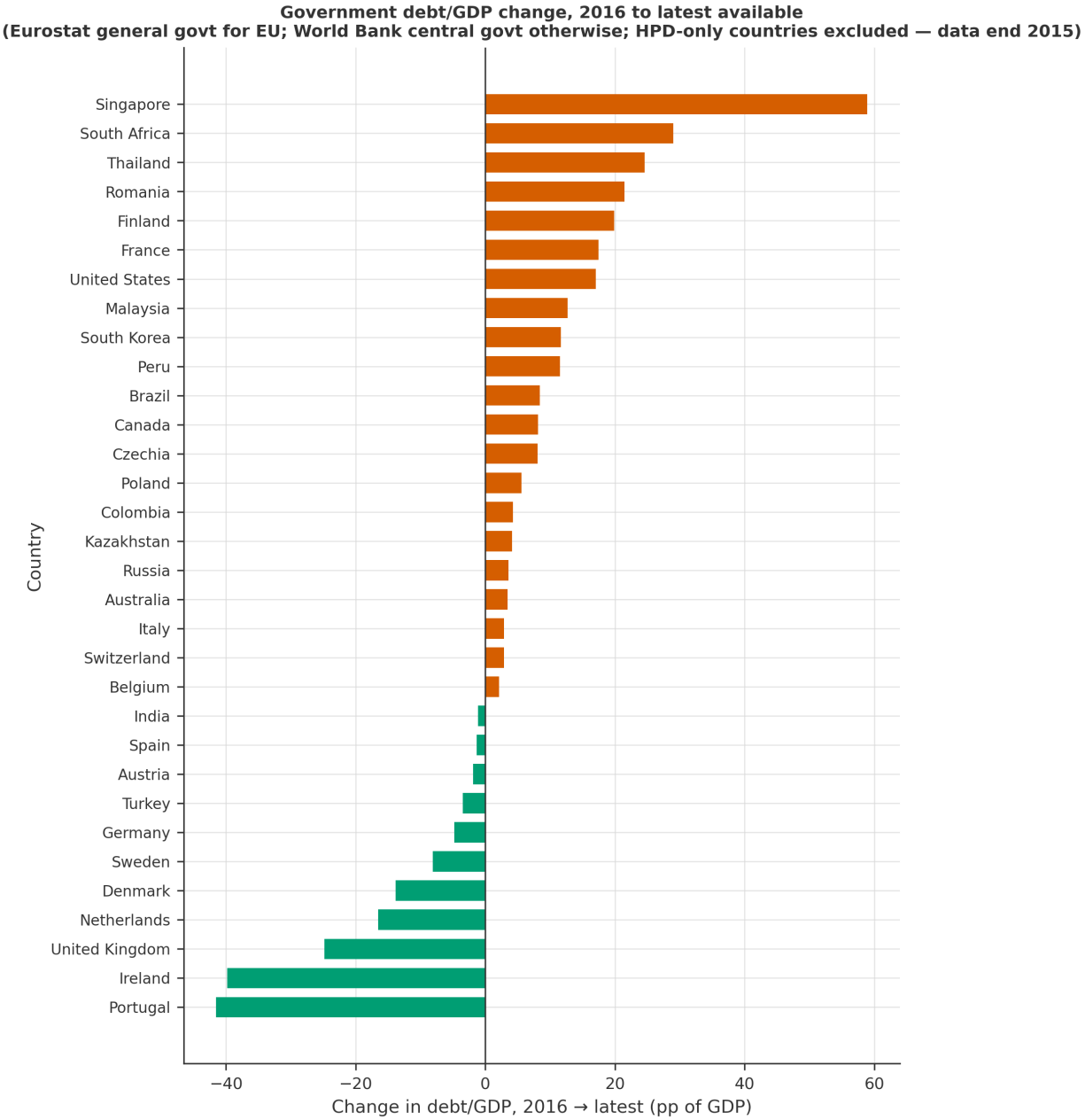


Figure 4: Change in government debt as percent of GDP, 2016 to latest available. Red: increase; green: decrease. *Source:* Eurostat general-government debt (EU economies); World Bank WDI central-government debt (others). Sector definitions differ across the two sources; HPD-only countries (debt data ending 2015) are excluded. Data vintage September 2026.

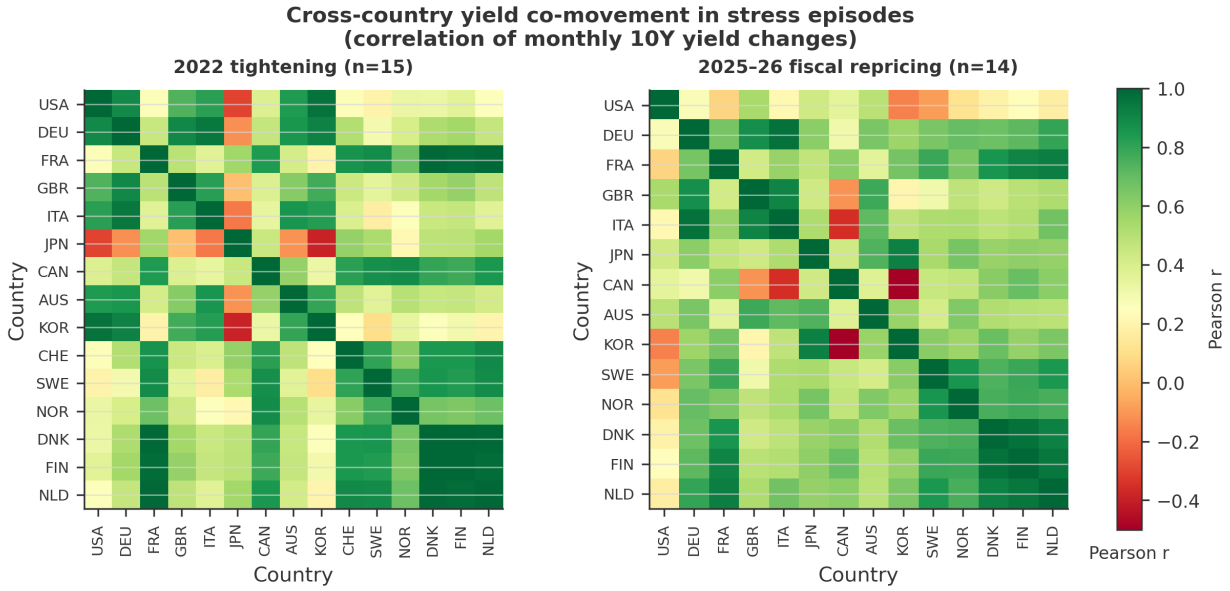


Figure 5: Pairwise Pearson correlations of monthly 10-year yield changes: the 2022 tightening (left) and the 2025–26 fiscal repricing (right). Darker green is higher correlation; red is negative. *Source:* author's calculations from the series in Figure 1.